

# T7 ARCHITECTURE

[Draw your reader in with an engaging abstract. It is typically a short summary of the document. When you're ready to add your content, just click here and start typing.]

Structure, Function and  
Performance

# T7 Architecture

## Structure, Function and Performance

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# T7 Architecture

Structure, Function and Performance

## System Architecture

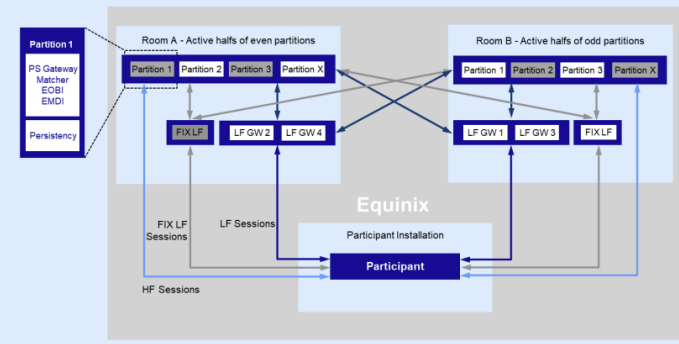
### T7 Topology Overview

## T7<sup>®</sup> Architecture Overview

T7<sup>®</sup> architecture developed by Deutsche Börse:

- Uses state-of-the-art infrastructure and hardware for high performance
- Offers reliable connectivity and enriched trading functionality
- Is multi-market capable, operates Derivatives (Eurex, EEX) and Cash markets (Xetra, Börse Frankfurt, Vienna, etc.)
- T7<sup>®</sup> consists of partitions. A partition is a failure domain in charge of matching, persisting and producing market data for a subset of products. Each T7 partition is distributed over two rooms in the Equinix data centre.
- There are 12 Eurex T7 and 11 Xetra T7 partitions.
- Separate partitions are used for markets of other exchanges hosted on T7 (e.g. Vienna (XVIE), EEX (XEEE), Bulgaria (XBUL), ...).
- The reference data contains the mapping of products to partition IDs.
- 4 LF gateways and one FIX LF gateway allow access to all Eurex partitions and the separate EEX partition.
- 4 LF gateways and one FIX LF gateway allow access to all Xetra partitions.
- 2 LF gateways and one FIX LF gateway are shared between Vienna and their partner exchanges.
- 2 LF gateways and one FIX LF gateway are shared between XBUL and XMAL

- Note that the active half of a partition is either on side A (for even partitions) or on side B (for odd partitions).
- In case of the failure of a PS gateway/Matching Engine or a market data publisher for EOBI or EMDI which is integrated into the Matching Engine, the active half of the service will shift to the other room.
- With consolidated PS gateway/Matcher process the active PS gateway and active Matching Engine act as a single failure domain within each partition, i.e. they will always fail as a single logical group.



# T7 Architecture

Structure, Function and Performance

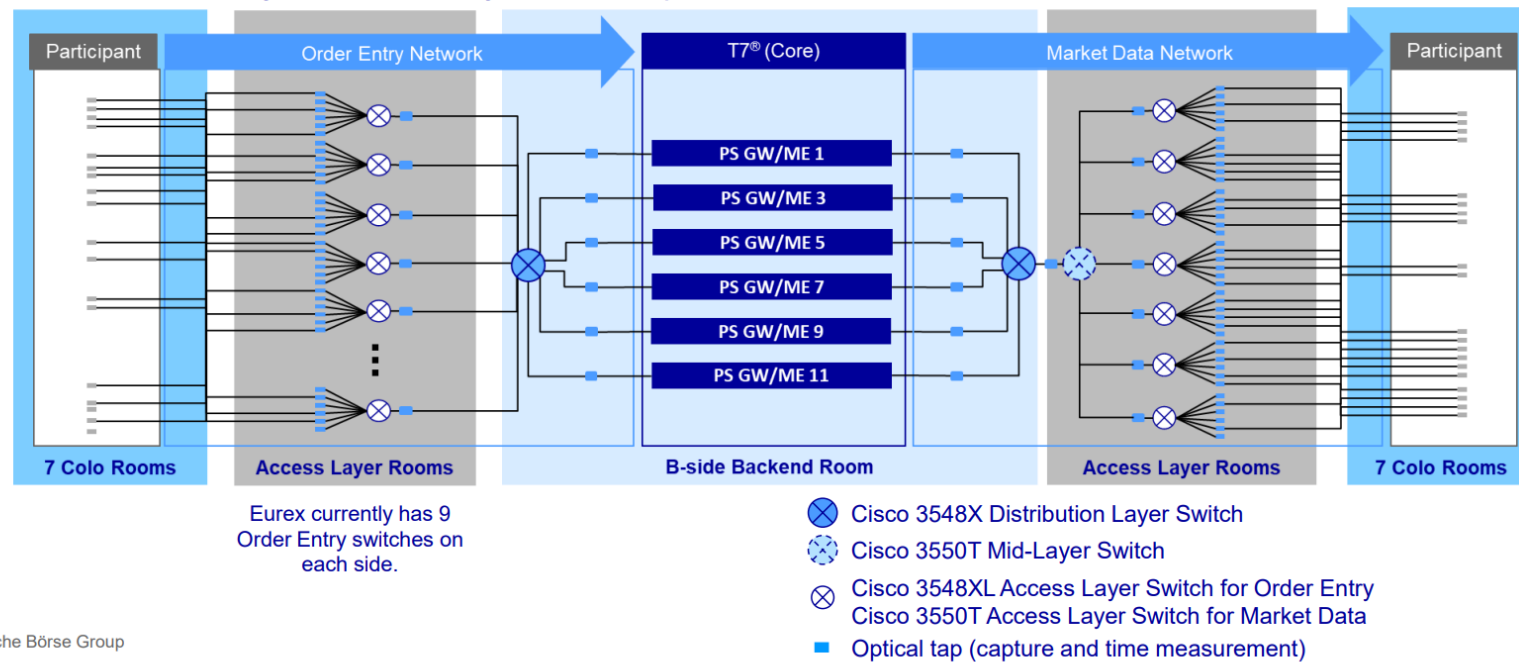
Topology- at a glance

## T7® Topology

At a glance

The below shows the Eurex B side (uneven partitions) as a schematic example of the topology of the T7 system.

Note that Xetra has only two Access Layer switches per side.

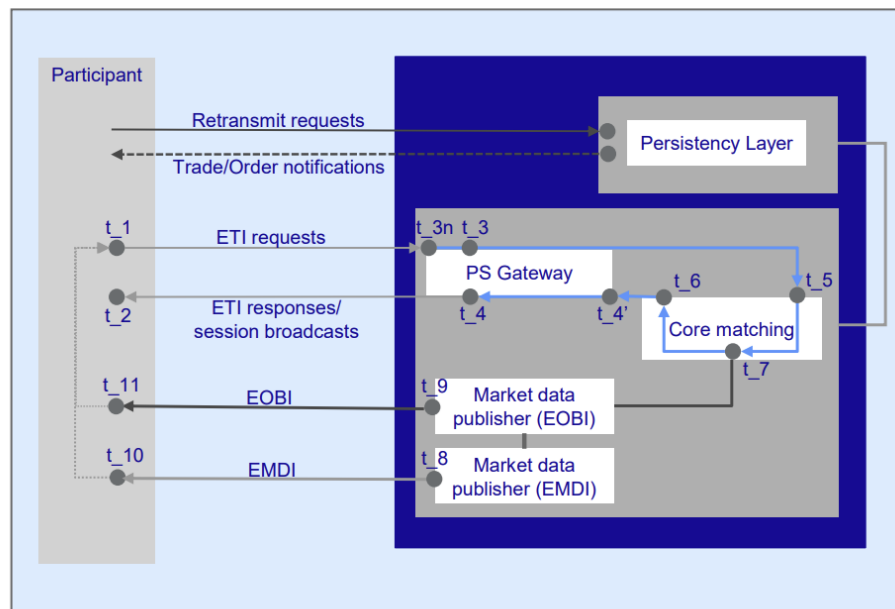


# T7 Architecture

## Structure, Function and Performance

### Topology – Overview

## T7® Topology Overview



### Matching Engine:

- order book maintenance & execution
- creation of direct responses as well as execution messages for passive orders/quotes
- creation of EOBI order book messages
- creation of EOBI order book snapshot messages
- creation of EMDI order book delta messages
- creation of EMDI order book snapshot messages

### Persistency:

- persistent order storage
- trade/execution history
- transaction history for standard orders
- creation of listener broadcast for standard orders

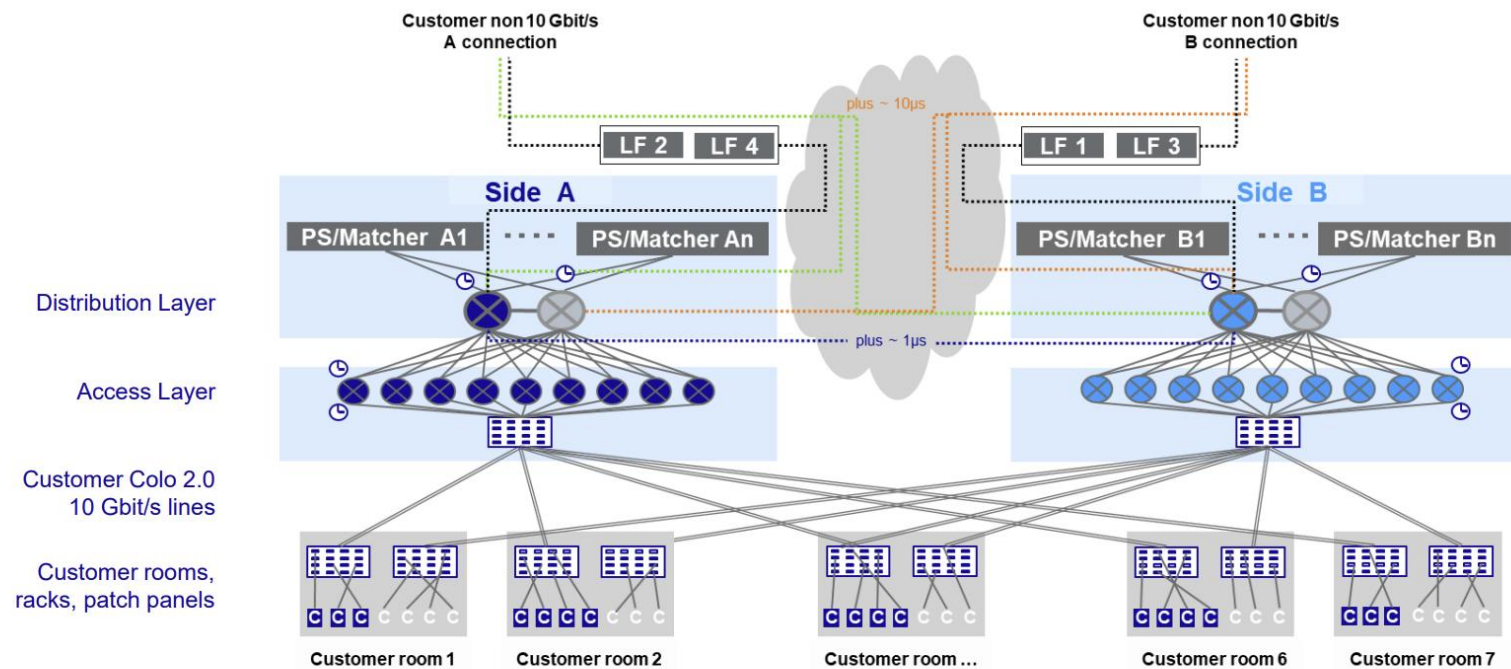
# T7 Architecture

Structure, Function and Performance

Topology- Co-Location

## Network Topology in Co-Location

Eurex Order Entry



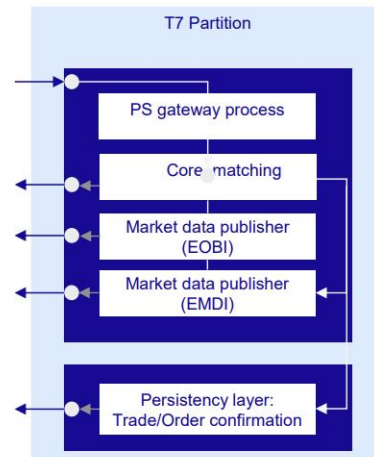
# T7 Architecture

## Structure, Function and Performance

### Partitions

## T7® Topology

### Partitions



Orders/quotes entered for a specific product are sent by the PS gateway process to the core Matching Engine process (both residing on the same server in the same partition).

The matching priority is assigned when the orders/quotes are read into the Matching Engine.

The core matching component works as follows:

- when an order/quote arrives, it is functionally processed (e.g. put in the book or matched),
- handover of data to the EOBI market data publisher, followed by EMDI market data publisher and
- handover of all data resulting from the (atomic) processing of the incoming order/quote to persistency component in the partition.

Resulting responses and private broadcasts are sent out in the following order:

- direct response to the order/quote entered (for persistent as well as for non-persistent orders and quotes),
- fast execution information for booked orders/quotes (in case of a match).

The generation of listener broadcasts, trade confirmations (by the persistency layer) and of non-EOBI/non-EMDI market data is done by separate processes. Hence the order of the resulting messages from these processes is not strictly deterministic.

**Note that the Matching Engine holds states of orders in memory. All responses, broadcasts, EOBI and EMDI market data thus are preliminary by nature.**



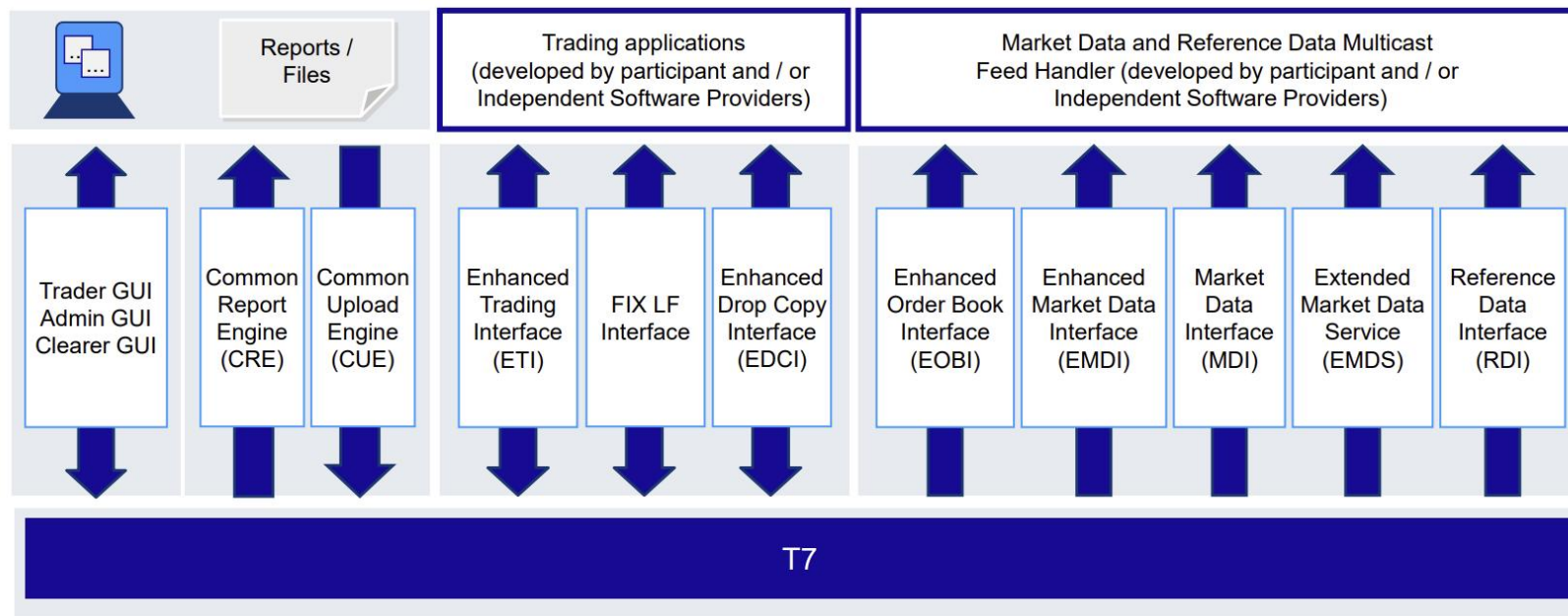
# T7 Architecture

Structure, Function and Performance

## Interfaces

### T7<sup>®</sup> Trading System

#### Interfaces



# T7 Architecture

## Structure, Function and Performance

### Order Entry Network

## Order Entry

### Introduction

Requests sent to T7 will be routed via an access network and a gateway.

There are the following basic connection alternatives:

#### Choice of Network

There are two network classes connecting a participant's installation with the T7 gateways:

- Co-Location with 10 Gbit bandwidth and a one-way base latency of around 2  $\mu$ s.
- Other networks with less than 1Gbit bandwidth and a one-way base latency of minimum about 50  $\mu$ s.

#### Choice of Session Type

T7 supports three session types:

- High frequency sessions connect to PS gateways for low latency access to a single partition (flat binary protocol: ETI). Please note that HF sessions can only be used within Equinix FR2 data center.
- Low frequency sessions connect to LF gateways for convenient access to all partitions, with a considerably higher base latency (ETI).
- FIX LF sessions connect to FIX LF gateway for convenient access to all partitions using the FIX protocol, with a considerably higher base latency than LF gateways.

#### Remarks

- LF gateway and FIX LF requests are routed via PS gateways.

# T7 Architecture

## Structure, Function and Performance

### Order Entry Gateways

There are three gateway types to access the T7 system:

#### **Partition-specific (PS) Gateway combined with Matching Engine**

Protocol: flat binary (ETI)  
Allowed session types: High Frequency Sessions only  
Sequencing: FIFO operation (Sequence guaranteed from network card to Matching Engine in)  
Latency: lowest, median latency ~ 12  $\mu$ s network card to Matching Engine in  
Versatility: Allows routing to one partition only, only subset of broadcasts available

#### **Low Frequency (LF) Gateway**

Protocol: flat binary (ETI)  
Allowed session types: Low Frequency Sessions only  
Sequencing: FIFO not guaranteed  
Latency: medium, (additional ~32  $\mu$ s latency compared to PS gateway direct access)  
Versatility: Routes to all partitions (via PS gateway), all ETI broadcast types available

#### **FIX LF Gateway**

Protocol: FIX  
Allowed session types: Fix Sessions only  
Sequencing: FIFO guaranteed  
Latency: high, requests to the Matching Engine are routed via PS gateways  
Versatility: Routes to all partitions (via PS gateway), all FIX broadcast types available

# T7 Architecture

Structure, Function and Performance

Market Data

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## Market Data

### Overview

Market Data can be consumed over two distinct types of networks and in various types

#### Choice of Network

There are two network classes available for market data:

- Co-Location with 10 Gbit bandwidth and a one-way base latency of around 2  $\mu$ s.  
10 Gbit connections are equalized in length (cable latency difference of less than  $\pm 1.75$  ns) and provide the lowest jitter.  
The reduction from  $\pm 2.5$  ns was achieved as an intermediate step and Deutsche Börse plans to further improve the cable length normalization.
- Other networks with less than 1Gbit bandwidth with higher base latency.

#### Choice of Market Data Type

There are three market data types:

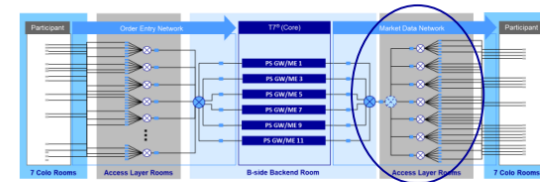
- Order by Order market data (EOBI) with highest granularity and lowest latency in flat binary format.  
EOBI is sent out directly from the Matching Engine and is only available via 10 Gbit network.
- Price level aggregated market data (EMDI) with slightly higher latency in FAST encoded format.
- Netted price level aggregated market data (MDI) in FAST encoded format.

# T7 Architecture

## Structure, Function and Performance

### Market Data Network

#### Cisco 3550T



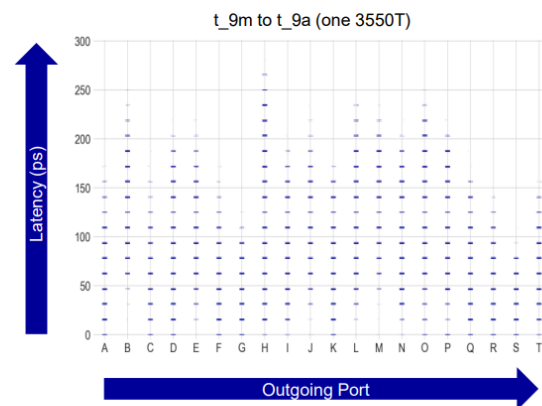
The graphs below show the processing times measured between  $t_{9m}$  and  $t_{9a}$  based on production data.

The first graph represents the per port processing time differences on one Eurex Access Layer switch normalized to the fastest port for each packet. The 15 ps steps between the clusters are due to the measurement device resolution.

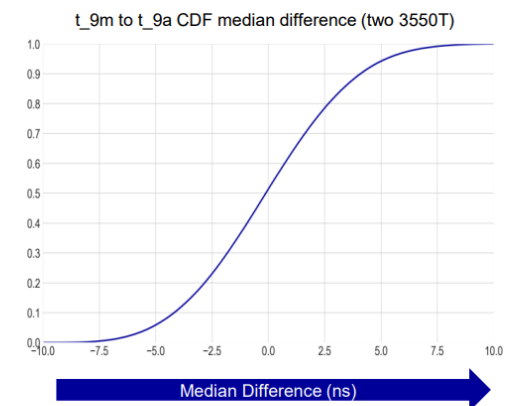
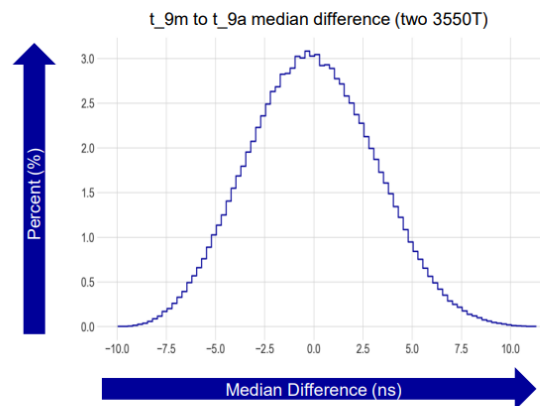
The second and third graphs show the difference in per switch median processing time between two Access Layer switches, in this case for Xetra as an example. Eurex and Xetra market data switches have a similar latency profile.

We observe the following behavior of the Cisco 3550T:

- Processing times within a single switch are consistently distributed within a range of ~300 ps across all ports after the latest firmware update.
- The median processing time differences between two switches follow a normal distribution with a higher variance than the processing time on a single switch.



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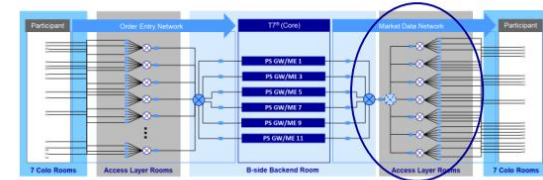
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# T7 Architecture

## Structure, Function and Performance

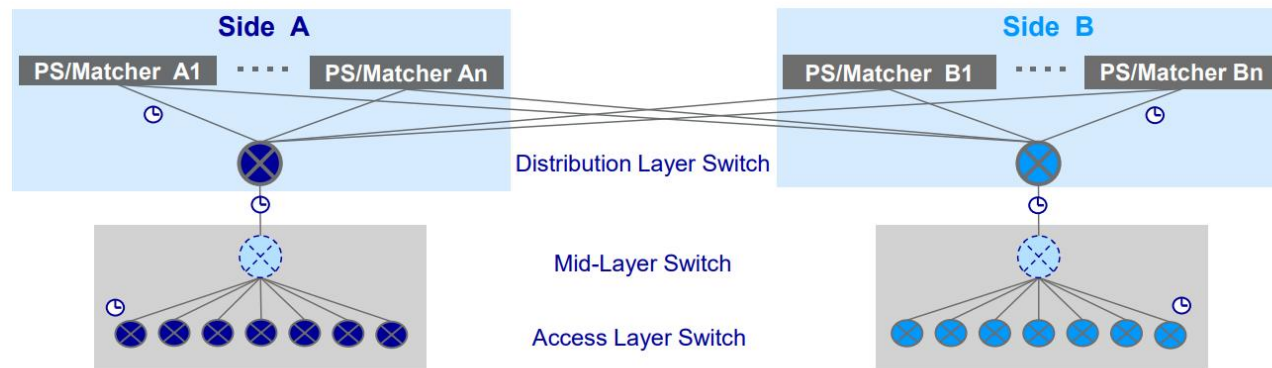
### Market Data Network

#### Mid-Layer Switch



Opposite to the Order Entry network, the Market Data network has an additional Mid-Layer Switch between the Distribution Layer and Access Layer switches for Eurex and Xetra. The purpose of the Mid-Layer is to:

- Enable equal cable length to Access Layer switches, this is achieved by physical proximity of Mid-Layer and Access Layer switches.
- Serialize the connection from T7 backend to Access Layer by having just one connection between Distribution Layer and Mid-Layer.
- Make use of the positive characteristics of the Cisco 3550T to achieve better distribution of market data to the Access Layer switches.





# T7 Architecture

Structure, Function and Performance

Middleware, Network, Hardware and OS Hardware

## Middleware, Network, Hardware and OS Overview

### **T7 uses state-of-the-art infrastructure components**

Intel(R) Xeon(R) Platinum 8462Y+ CPUs for all servers.

We currently use Red Hat Enterprise Linux 8.10 and plan to migrate to version 9.0 mid 2025.

T7 internal communication between its core components is based on Ethernet.

### **T7 network access**

Deutsche Börse offers trading participants to connect via 10 Gbit/s cross connects to its T7 platform in the Equinix data centre.

The Co-Location offering uses Cisco Nexus 3548-XL switches for Order Entry and is operating in cut-through mode. For market data Cisco Nexus 3550-T are used.

All cables are normalized to give an overall maximum deviation between any two cross connects of less than  $\pm 0.35$  m ( $\pm 1.75$  ns).

The reduction from  $\pm 0.5$  m was achieved as an intermediate step and Deutsche Börse plans to further improve the cable length normalization.

Insight into network dynamics is offered by the High Precision Timestamp File service (see <https://www.mds.deutsche-boerse.com/mds-en/analytics/high-precision-timestamps>).

Participant facing interface cards on the gateways and market data publishers use Solarflare EnterpriseOnload wire order delivery API to bypass the kernel TCP stack and deliver messages in the same order received by the network card.

Cables connecting Line-of-Sight antenna cables have been equalized to  $\pm 1$  m by Equinix.

# T7 Architecture

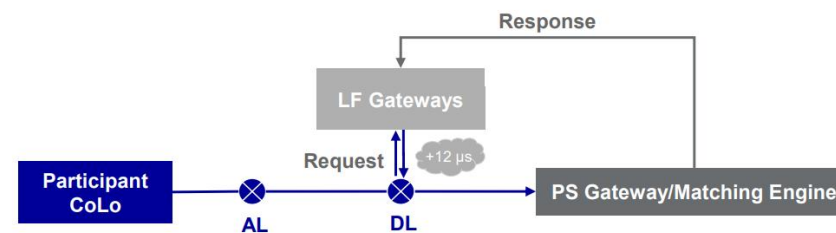
Structure, Function and Performance

## Performance

### PS Gateway vs LF Gateways

#### PS Gateway versus LF Gateways

Latency comparison



- Using 10 Gbit/s cross connects and access via PS gateways provides the fastest way for order and quote management in T7.
- LF gateways on the other hand allow access to all partitions of a market via a single session.
- Some markets (e.g. XMAL) are using LF gateways only.
- Most markets have combined PS gateway/ Matching Engines in place for which all Matching Engine bound requests sent to LF gateways are routed via PS gateways.
- The base inbound latency for LF gateway transactions is around 12 μs higher compared to directly accessing the PS gateways. Note also that requests that have to cross sides between LF and PS gateways take another ~1 μs to reach the Matching Engine. This was significantly reduced from ~45 μs to ~1 μs end of January 2025.



# T7 Architecture

## Structure, Function and Performance

### Throttle and Session Limits

## Throttle and Session Limits

**In order to protect the trading system, T7 has several measures in place to ensure that its most vital components are not harmed by a malfunctioning client application. Therefore, transaction limits are imposed on T7 sessions.**

ETI LF sessions are available with throttle values of 150 or 50 transactions/sec. ETI HF sessions are available with throttle values of 250\*, 150 or 50 transactions/sec. Furthermore, LF sessions that cannot enter orders/quotes but can only receive trade and listener broadcasts are available (at a reduced price).

The disconnect limit is set at:

- 750 for HF Ultra sessions with a throttle value of 250 transactions/sec. i.e. a session will be disconnected in case of more than 750 consecutive rejects due to exceeding the transaction limit (throttle).
- 450 for sessions with a throttle value of 150 transactions/sec, i.e. a session will be disconnected in case of more than 450 consecutive rejects due to exceeding the transaction limit (throttle).
- 150 for sessions with a throttle value of 50 transactions/sec, i.e. a session will be disconnected in case of more than 150 consecutive rejects due to exceeding the transaction limit (throttle).

Please note that in case the disaster recovery facility is used, all ETI sessions will have a throttle limit of 30 transactions per second.

For both limits, all technical transactions are counted using a sliding window.

The number of ETI sessions which can be ordered is limited. Currently, up to [600](#) Eurex sessions and 200 Xetra sessions per participant can be ordered.

There is also a limit on the maximum number of sessions that can connect to a PS gateway concurrently per participant. This limit is currently configured to 80 sessions, see [Eurex Circular 122/17](#).

On 1 July 2019, we introduced a limit on the maximum number of outstanding session and trader login requests possible per business unit and per session at any given point in time. This limit is set to 50 on business unit level, 10 on session level. We recommend a synchronous login procedure, where a login request is sent on a session only after the previous login has been responded to. Please refer to the Incident Handling Guide for details.

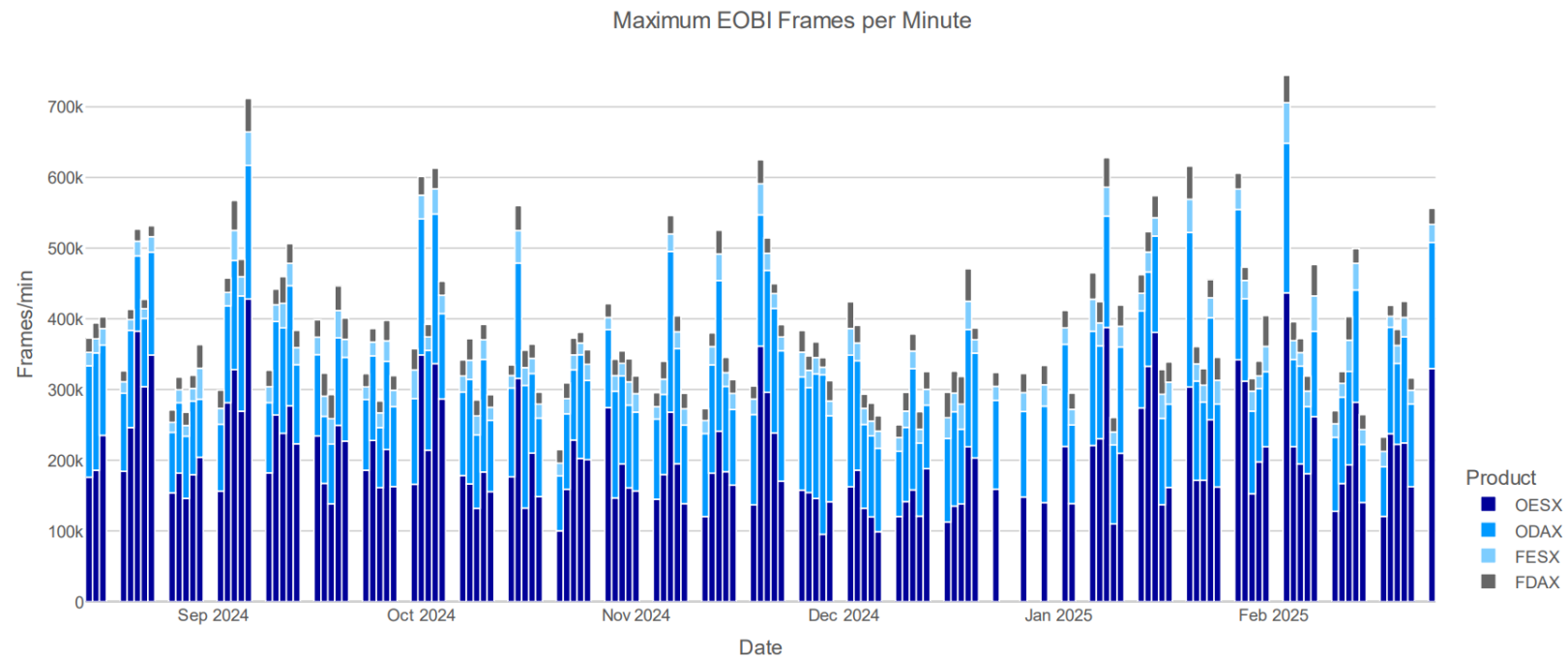
For Eurex the number of order entry cross connects in colocation that may be used concurrently on a single day is limited to 6 per Access Layer switch. In addition, the number of allowed ethernet frames per cross connect is currently limited to [30.000](#) per second and [600.000](#) per minute for Eurex and 25,000 per second and [600.000](#) per minute for Xetra.

# T7 Architecture

Structure, Function and Performance

EOBI – Maximum Frames Per Minute

## Maximum Frames per Minute - EOBI



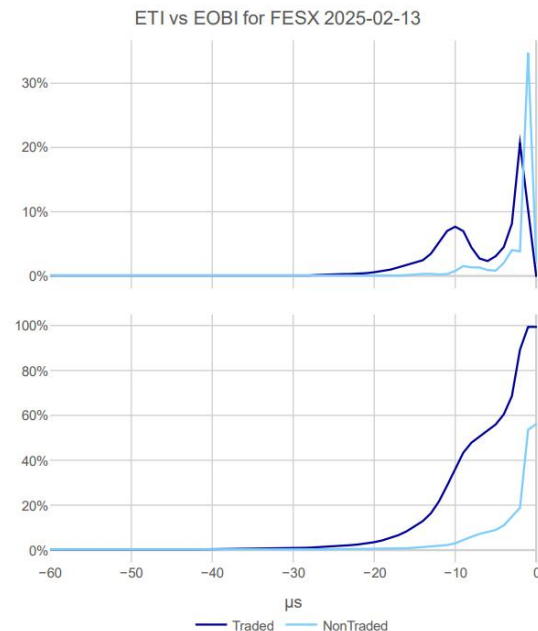
# T7 Architecture

Structure, Function and Performance

Trading System Dynamics – Latency Characteristics of EOBI vs ETI for Futures

## Trading System Dynamics

Latency characteristics of EOBI versus ETI for Futures



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T7 is designed to publish order book updates first on its public data feed.

The diagram shows the time difference distribution between public and private data in microseconds (EOBI first datagram vs ETI responses,  $t_9 - t_4$ ).

The data is a production sample from 13 February 2025.

EOBI market data is in median 1.5  $\mu\text{s}$  faster than the ETI response for order book updates and 7.5  $\mu\text{s}$  faster for executions.

The first EOBI datagram is faster in more than 99.4 percent of the cases compared to the ETI response, and also the first passive ETI book order notification for simple transactions.

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# T7 Architecture

## Structure, Function and Performance

### Order Entry Latency – Comparison of Access Types

#### Order Entry Latency

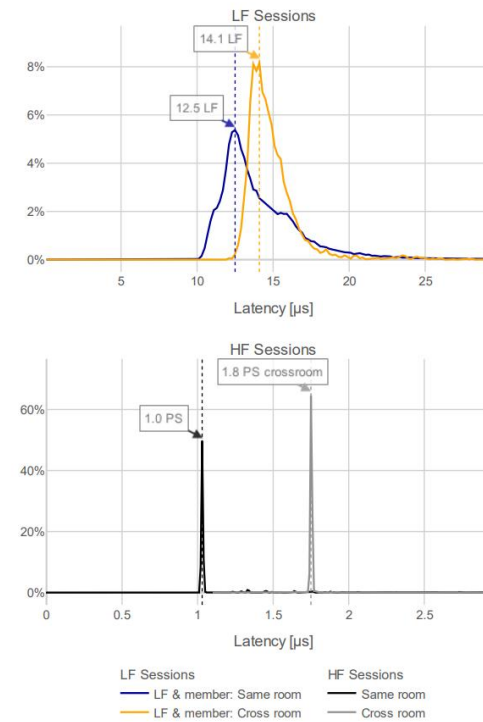
#### Comparison of access types

The table below shows a comparison of different access options to the T7 system. All times given are in microseconds.

Network timestamps (t\_3a) are synchronized using white rabbit. The time synchrony quality between these timestamps is thus ~1 ns. Other T7 timestamps are subject to jitter of up to ±50 ns.

| Gateway type | Same side line                      | Same side partition                 | t_3a to t_3n uncongested latency |
|--------------|-------------------------------------|-------------------------------------|----------------------------------|
| PS           | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | 1                                |
|              | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | 2                                |
| LF           | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | 12                               |
|              | <input checked="" type="checkbox"/> | <input type="checkbox"/>            | 13                               |
|              | <input type="checkbox"/>            | <input checked="" type="checkbox"/> | 13                               |
|              | <input type="checkbox"/>            | <input type="checkbox"/>            | 14                               |

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# T7 Architecture

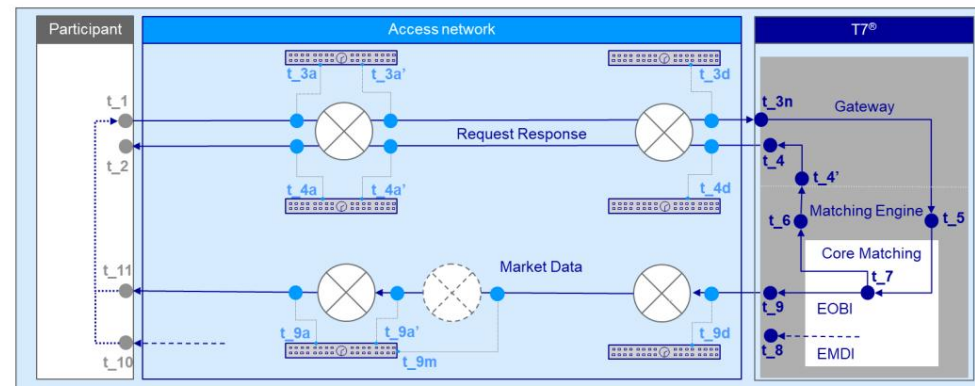
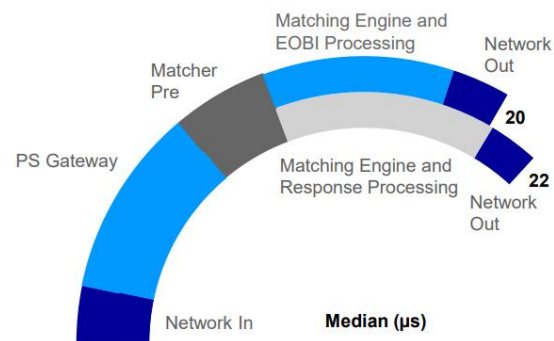
## Structure, Function and Performance

### Order Entry Network – Latency Composition

## T7<sup>®</sup> Latency Composition

The T7 trading system provides utmost transparency about its latency characteristics. Most of the timestamps taken at the various measurement points within T7 Core are included in each ETI response and EOBI market data. With the high precision timestamp file, we also make three network timestamps available for each EOBI market data packet ( $t_{9d}$ ) and its triggering transaction ( $t_{3a}$ ,  $t_{3d}$ ).

The latency circle shows the median latencies for the request-response/EOBI market data path for Eurex futures sent via high frequency sessions measured on 13 February 2025.



# T7 Architecture

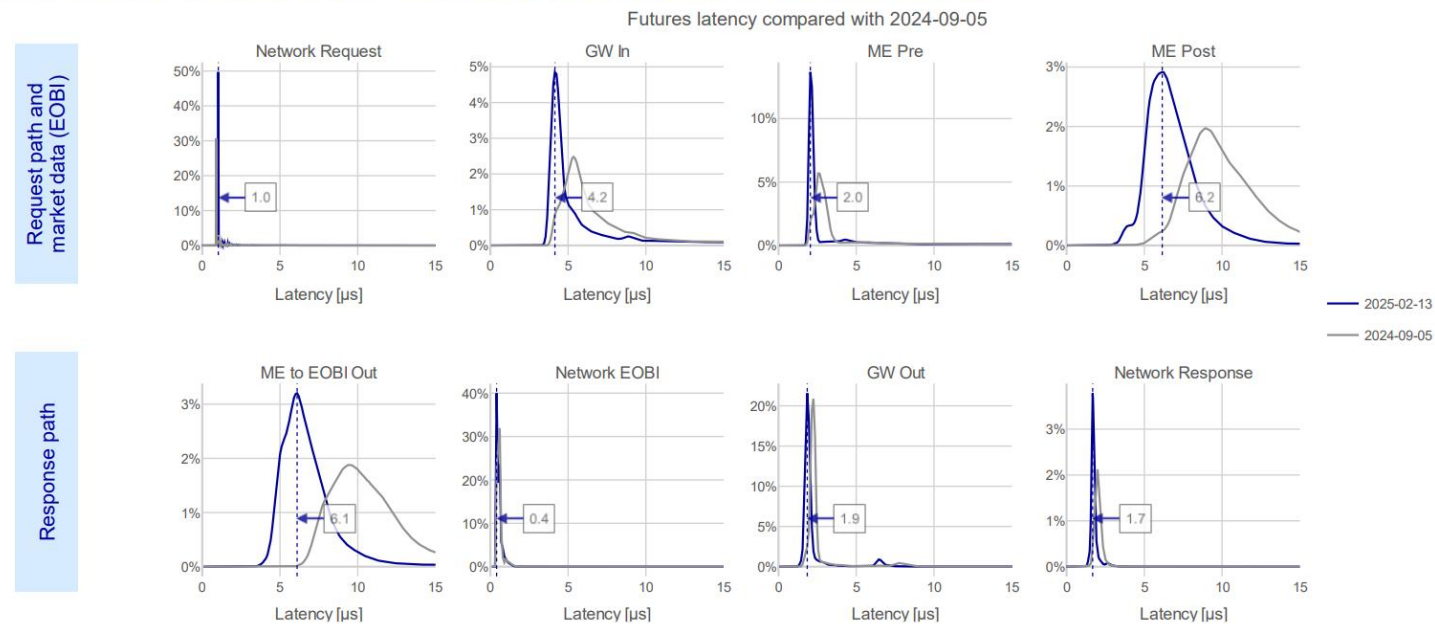
## Structure, Function and Performance

### Order Entry Network – Latency Composition – Continued

## T7<sup>®</sup> Latency

### Composition (continued)

The below charts show a comparison of current latencies with the spectrum from before the technical upgrade of T7 core trading system. The data is for Eurex Futures sent from HF sessions. 'Network response/EOBI' include the TCP/UDP stack.



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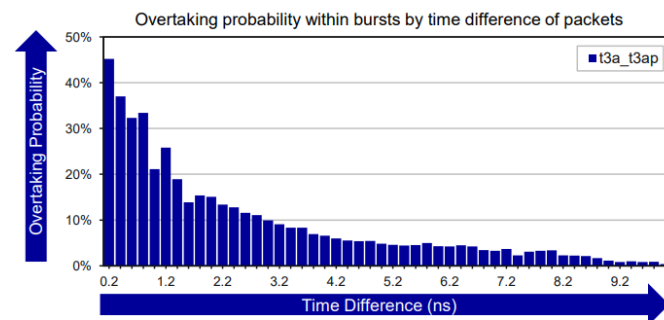
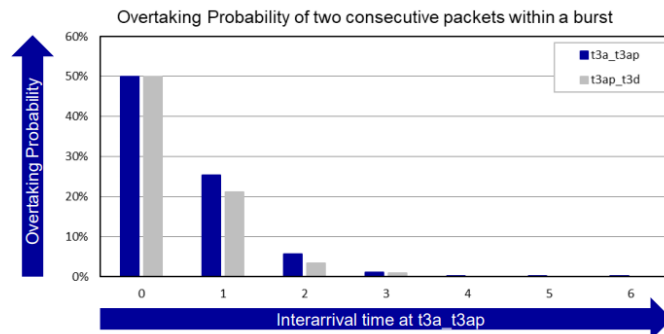
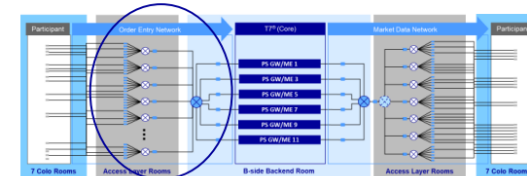
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# T7 Architecture

## Structure, Function and Performance

### Order Entry Network – Latency Aspects

## Order Entry Network Latency aspects

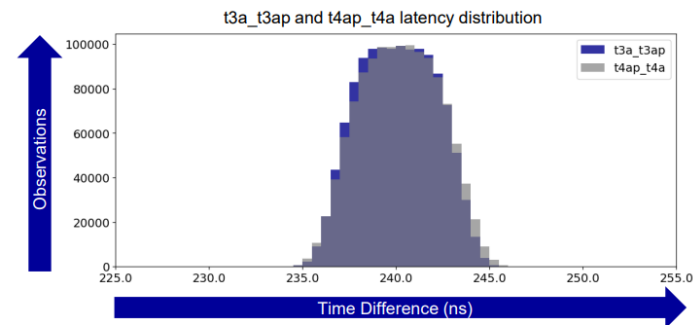


The chart on the left shows the overtaking probability between a first and a second message within a burst on one switch. The overtaking drops sharply and there is no observed overtaking beyond 3 nanoseconds delta between messages.

With improved timestamp accuracy and introduction of sub-nanoseconds on capture devices, more granular analyses could be performed. The bottom left graph shows the overtaking probability of all messages within a burst based on their interarrival time.

The base latency and latency jitter is identical for all Access Layer switches within the measurement precision (bottom right graph).

Data is taken from the Eurex A side.





# T7 Architecture

Structure, Function and Performance

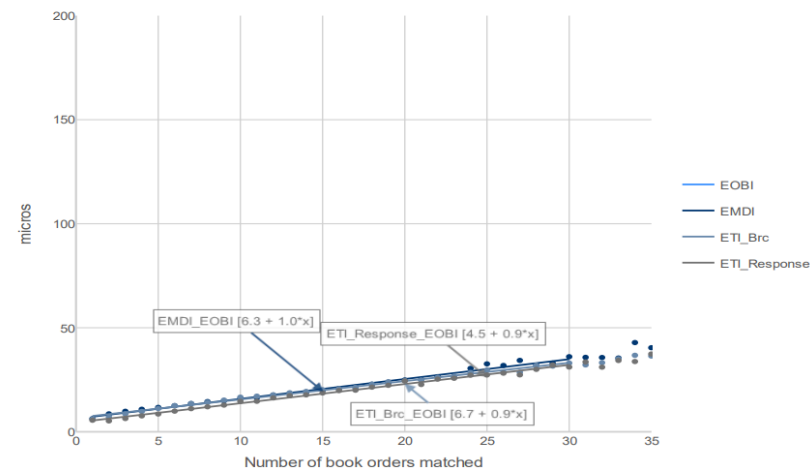
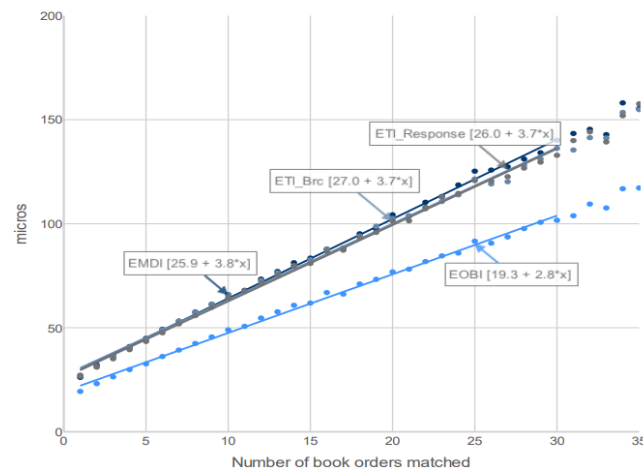
Latency characteristics of ETI versus EOBI versus EMDI

## Trading System Dynamics

### Latency characteristics of ETI versus EOBI versus EMDI

The diagrams below display the dependency of the median latency on the complexity of a trade for ETI (the right diagram is normalized to EOBI) ( $t_4 - t_7$ ), EMDI ( $t_8 - t_7$ ) and EOBI ( $t_9 - t_7$ ). Note that for ETI we display the gateway sending time of the first passive notification and for EOBI the sending time of the UDP datagram containing the Execution Summary message.

The difference between public and private data has slightly been narrowed with the integration of EMDI into Matching Engine process. However, the 'public data first' principle is still being ensured. In over 99% of all trades, we disseminate order book data on EOBI first (also for larger trades).



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# T7 Architecture

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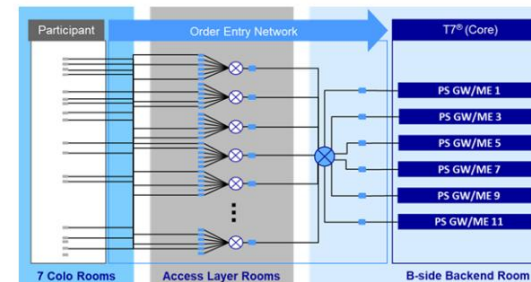
Order Entry Network – Colocation 2.0

## T7<sup>®</sup> Latency

### Order entry network

The order entry network in Colocation 2.0 uses a two staged hierarchical funnel in approach.

All cables are normalized to guarantee a maximum deviation of  $\pm 0.35$  m ( $\pm 1.75$  ns) between any two cross connects to the exchange. The reduction from  $\pm 0.5$  m was achieved as an intermediate step and Deutsche Börse plans to further improve the cable length normalization.



Every inbound and outbound packet on this path is captured via passive network TAPs at 3 different locations.

The packets are timestamped with sub-nanosecond resolution and nanosecond accuracy.

This capture infrastructure allows early detection and analysis of many kinds of technical network problems and an in-depth latency analysis on network level, like overtaking probabilities between measurement points.

The high precision timestamp file service enables participants access to timestamps  $t_{3a}$ ,  $t_{3d}$  and  $t_{9d}$  for each request leading to an EOB market data update.

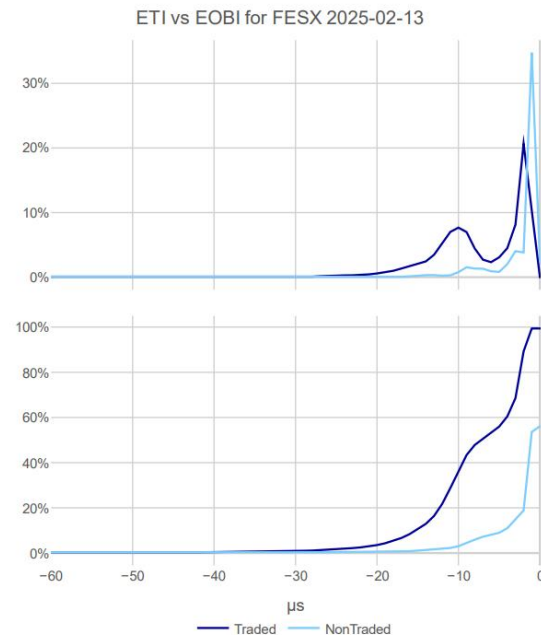
# T7 Architecture

Structure, Function and Performance

## Latency Characteristics of EOBI vs ETI for Futures

### Trading System Dynamics

#### Latency characteristics of EOBI versus ETI for Futures



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The diagram shows the time difference distribution between public and private data in microseconds (EOBI first datagram vs ETI responses,  $t_9 - t_4$ ).

The data is a production sample from 13 February 2025.

EOBI market data is in median 1.5 µs faster than the ETI response for order book updates and 7.5 µs faster for executions.

The first EOBI datagram is faster in more than 99.4 percent of the cases compared to the ETI response, and also the first passive ETI book order notification for simple transactions.

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